Measure and Probability on Finite Sets

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Despite its infamous reputation, the foundations of probability theory are straightforward. Much of the mathematical distress arises only when we implement probability theory on elaborate sets like the real numbers. Similarly, most of the philosophical awkwardness arises only when trying to assign an interpretation to the mathematics. In this chapter we will avoid this baggage by introducing the basics of abstract probability theory on the simplest nontrivial set: a collection of a finite number of elements.

1 Finite Sets

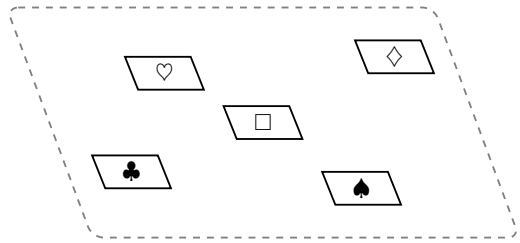
A **finite set** contains a finite number of distinct elements,

$$X = \{x_1, ..., x_N\}.$$

The numerical index here allows us to differentiate the between the N individual elements, but it does not necessarily imply that the elements have any particular ordering to them. To avoid any assumption of ordering, I'll use the five element set (Figure 1)

$$X = \{\square, \clubsuit, \diamondsuit, \heartsuit, \spadesuit\}$$

for demonstration.



$X = \{\square, \clubsuit, \diamondsuit, \heartsuit, \spadesuit\}$

Figure 1: A finite set contains a finite number of elements. This particular set contains five elements.

In practical applications of probability theory, the abstract elements x_n will model meaningful objects. In this chapter, however, I will avoid any particular interpretation and instead focus on the basic mathematical concepts. That said, when X is intended to capture *all* of the objects of interest in a given application I will refer to it as the **ambient set**.

Once we've defined an ambient set, we have various ways of organizing its individual elements and manipulating those organizations.

1.1 Subsets

A **subset** of X is any collection of elements in X. To avoid any ambiguity, I will exclusively use lowercase roman letters x to denote a variable element in the ambient set X and lowercase sans serif letters x to denote a variable subset. For example, $\mathsf{x} = \{\square, \diamondsuit, \heartsuit\}$ is a subset of $X = \{\square, \clubsuit, \diamondsuit, \heartsuit, \spadesuit\}$ (Figure 2).

Importantly, a subset implies a notion of *membership* but not a notion of *multiplicity*. While a subset might include an element x_n, it cannot include it multiple times.

If x is a subset of the ambient set X, then we write $\mathsf{x} \subset X$. When x might contain all of the elements of X, in which case $\mathsf{x} = X$, then we write $\mathsf{x} \subseteq X$. Some authors use $\subset$ to refer to both circumstances, but I prefer the more precise notation that distinguishes between the two.

Subsets are *recursive* objects. Selecting any elements from a subset yields a new subset. For example the collection $\{\diamondsuit, \heartsuit\}$ is a subset of both the previously introduced subset,

$$\{\diamondsuit, \heartsuit\} \subset \{\square, \diamondsuit, \heartsuit\},$$

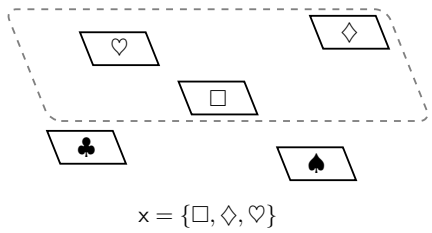


Figure 2: A subset $\mathsf{x} \subset X$ is any collection of elements from the ambient set X. Here $\mathsf{x} = \{\square, \diamondsuit, \heartsuit\}$ contains just three of the five elements in $X = \{\square, \clubsuit, \diamondsuit, \heartsuit, \spadesuit\}$.

as well as the full ambient set,

$$\{\diamondsuit, \heartsuit\} \subset X.$$

More formally, a subset x' is a subset of another subset x if and only if all of the elements in x' are also in x.

Regardless of how many elements a finite set X contains, we can always construct three special types of subsets. The **empty set** $\emptyset = \{\}$ contains no elements at all. On the other hand, the entire set itself can be considered a subset containing all of the elements. Lastly, a subset containing a single element is denoted $\{x_n\}$ and referred to as an **atomic set**.

There are

$$\binom{N}{n} = \frac{N!}{n!(N-n)!}$$

ways to select n elements from a finite set of N total elements, and hence $\binom{N}{n}$ total subsets of size n. For example there is only one subset that contains no elements,

$$\binom{N}{0} = \frac{N!}{0!(N-0)!} = \frac{N!}{N!} = 1,$$

which is just the empty set. Similarly there is only one subset that contains all of the elements,

$$\binom{N}{N} = \frac{N!}{N!(N-N)!} = \frac{N!}{N!} = 1,$$

which is just the full set itself. On the other hand there are

$$\binom{N}{1} = \frac{N!}{1!(N-1)!} = N$$

distinct atomic sets that contain a single element, one for each element in X.

Counting all of the subsets of all possible sizes gives

$$\sum_{n=0}^{N} \binom{N}{n} = 2^N$$

possible subsets that we can construct from a finite set with N elements. Those familiar with computer science might recognize that this is equivalent to needing N bits to specify each distinct subset, one bit for the inclusion or exclusion of each element.

The collection of all subsets is itself a finite set with 2^N elements. We refer to this set as the **power set** of X and denote it 2^X.

When the ambient set, and hence all of the possible elements that any subset might contain, is fixed, the prefix "sub" is sometimes dropped. In this case all subsets are simply referred to as sets. For example, collections of elements might be denoted "sets" when one takes the ambient set for granted and "subsets" when one wants to explicitly acknowledge the context of a particular ambient set. Precise terminology, however, can vary strongly between disciplines and even individuals.

1.2 Subset Operations

We can always construct subsets element by element. That said, we can also construct them by manipulating existing subsets.

For example, given a subset $\mathsf{x} \subset X$ we can construct its **complement** by collecting all of the elements in X that are not already in x. The atomic set $\mathsf{x} = \{\diamondsuit\}$ contains the lone element $\diamondsuit$ and its complement contains the remaining elements (Figure 3)

$$\mathsf{x}^c = \{\square, \clubsuit, \heartsuit, \spadesuit\}.$$

By construction, the complement of the empty set is the entire set, $\emptyset^c = X$, and the complement of the full set is the empty set, $X^c = \emptyset$.

More formally, the construction of complementary subsets defines a *unary* operation that takes in a subset as input and outputs the complementary subset,

$$\begin{aligned} \cdot^c : 2^X &\to 2^X \\ \mathsf{x} &\mapsto \mathsf{x}^c \ . \end{aligned}$$

The top line of this notation denotes the collection of all possible inputs and outputs, here both the power set, while the bottom line denotes the action on a particular input, here a subset mapped into its complement. For instance, applying the complement operator to the subset $\mathsf{x} = \{\clubsuit, \spadesuit\}$ gives

$$\mathsf{x}^c = \{\clubsuit, \spadesuit\}^c = \{\square, \diamondsuit, \heartsuit\}.$$

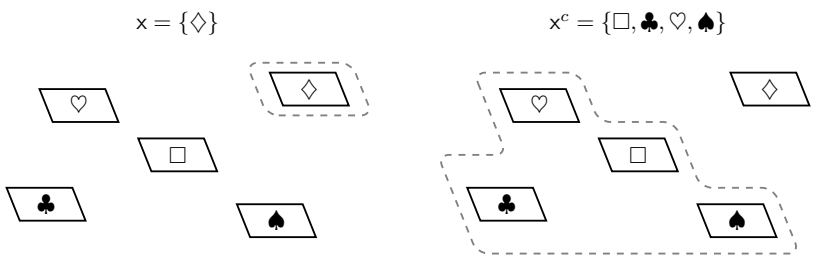


Figure 3: The complement of a subset x is the subset x^c consisting of all elements in the ambient set that are not in x.

We can also construct new subsets from two existing subset. Consider, for example, two subsets $\mathsf{x}_1 = \{\square, \heartsuit\}$ and $\mathsf{x}_2 = \{\square, \spadesuit\}$ (Figure 4). The collection of all elements that are contained in *either* subset is itself a subset,

$$\{\square, \heartsuit, \spadesuit\} \subset X,$$

as is the collection of all elements that are contained in *both* subsets,

$$\{\square\} \subset X.$$

These derived subsets are referred to as the **union**,

$$\mathsf{x}_1 \cup \mathsf{x}_2 = \{\square, \heartsuit\} \cup \{\square, \spadesuit\} = \{\square, \heartsuit, \spadesuit\},$$

and **intersection**,

$$\mathsf{x}_1 \cap \mathsf{x}_2 = \{\square, \heartsuit\} \cap \{\square, \spadesuit\} = \{\square\},$$

respectively (Figure 5).

The union and intersection operations are both symmetric in the sense that either ordering of the two input subsets results in the same output subset,

$$\mathsf{x}_1 \cup \mathsf{x}_2 = \mathsf{x}_2 \cup \mathsf{x}_1$$

and

$$\mathsf{x}_1 \cap \mathsf{x}_2 = \mathsf{x}_2 \cap \mathsf{x}_1.$$

Two subsets are **disjoint** if they don't share any elements. In this case, their intersection is the empty set,

$$\mathsf{x}_1 \cap \mathsf{x}_2 = \emptyset.$$

Because the empty set does not contain any elements, its union with any subset returns back that subset,

$$\mathsf{x} \cup \emptyset = \emptyset \cup \mathsf{x} = \mathsf{x},$$

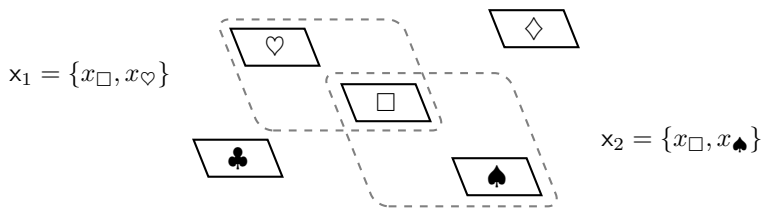


Figure 4: Two existing subsets can always be manipulated into two new subsets.

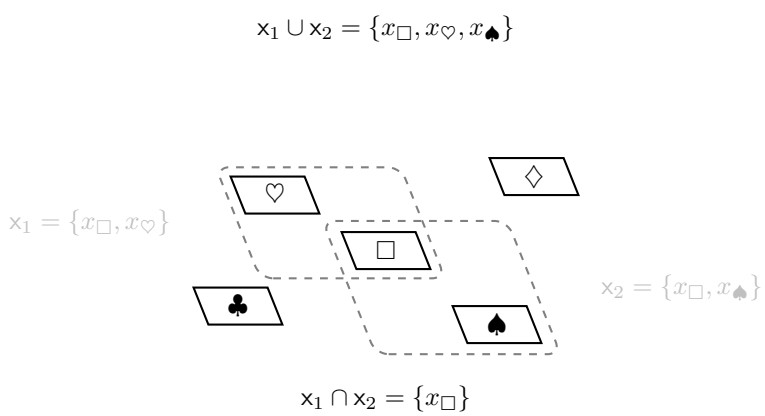


Figure 5: The union of two subsets, $\mathsf{x}_1 \cup \mathsf{x}_2$, is a subset containing all of the elements in either input subset. On the other hand, the intersection of two subsets, $\mathsf{x}_1 \cap \mathsf{x}_2$, is a subset containing just the elements that occur in both input subsets.

and its intersection with any subset returns the empty set,

$$\mathsf{x} \cap \emptyset = \emptyset \cap \mathsf{x} = \emptyset.$$

The union and intersection of a subset with itself always returns that subset,

$$\mathsf{x} \cup \mathsf{x} = \mathsf{x} \cap \mathsf{x} = \mathsf{x}.$$

Similarly, the union of any subset with the full set returns the full set,

$$\mathsf{x} \cup X = X \cup \mathsf{x} = X,$$

and the intersection of any subset with the full set returns back the subset,

$$\mathsf{x} \cap X = X \cap \mathsf{x} = \mathsf{x}.$$

Because they require two inputs, the union and intersection define *binary* operations that consume *two* input subsets and return a single output subset,

$$\begin{aligned} \cdot \cup \cdot : 2^X \times 2^X &\to 2^X \\ \mathsf{x}_1, \mathsf{x}_2 \quad &\mapsto \mathsf{x}_1 \cup \mathsf{x}_2 \end{aligned}$$

and

$$\begin{aligned} \cdot \cap \cdot : 2^X \times 2^X &\to 2^X \\ \mathsf{x}_1, \mathsf{x}_2 \quad &\mapsto \mathsf{x}_1 \cap \mathsf{x}_2. \end{aligned}$$

Here $2^X \times 2^X$ denotes the collection of all *pairs* of subsets.

2 Measure and Probability Over Elements

Measure theory, and its special case of probability theory, is often burdened with intricate if not mysterious interpretations. From a mathematical perspective, however, measure theory simply concerns the consistent **allocation** of some abstract quantity across an ambient set.

Consider a *reservoir* of some positive, real-valued, and *conserved* quantity, $M \in [0, \infty]$ (Figure 6). Because M is conserved, any amount m_n that is allocated to the element $x_n \in X$ has to be *depleted* from the total reservoir, leaving less to be allocated to the remaining elements.

We have to be careful if the total content of the reservoir M is infinite. In this case we can allocate an infinite amount from the reservoir while still having an infinite quantify left. At the same time, allocating an infinite amount can deplete the reservoir completely, but it can also leave any finite quantity. Infinity is, at the very least, mathematically subtle.

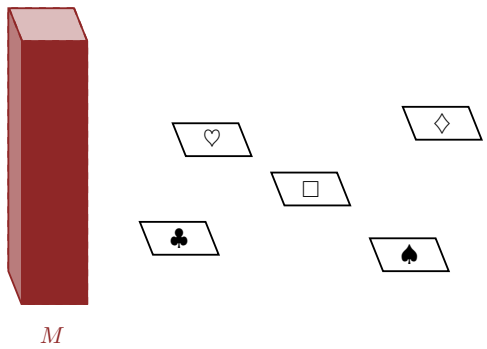


Figure 6: Measure theory concerns the allocation of some positive real-valued quantity M over the individual elements of the ambient set.

To make the mathematics as useful as possible, we will avoid endowing M with any particular meaning for the time being. Instead interpretations will arise only when we use these allocations to *model* meaningful systems.

An *exhaustive* allocation of M across the ambient set ensures that the reservoir is completely emptied. In another words, all of M has to be allocated to the elements $x_n \in X$.

For example consider our demonstrative ambient set $X = \{\square, \clubsuit, \diamondsuit, \heartsuit, \spadesuit\}$. Allocating $m_\square$ to $\square$ leaves $M - m_\square$ remaining to allocate to the other four elements (Figure 7a). Allocating $m_\clubsuit$ to $\clubsuit$ depletes the reservoir a bit more (Figure 7b). At the end we have to allocate all

$$M - m_\square - m_\clubsuit - m_\diamondsuit - m_\heartsuit$$

that remains to $\spadesuit$ to completely empty the reservoir (Figure 7e).

A **measure** is any consistent allocation of the quantity M to the elements of an ambient set. Mathematically, any measure over a finite set can be characterized by N positive real numbers (Figure 8)

$$\mu = \{m_1, \dots, m_N\}$$

that satisfy

$$0 \leq m_n$$

and

$$\sum_{n=1}^{N} m_n = M.$$

For example, any measure over the five-element set $X = \{\square, \clubsuit, \diamondsuit, \heartsuit, \spadesuit\}$ is specified by any five positive real numbers $\{m_\square, m_\clubsuit, m_\diamondsuit, m_\heartsuit, m_\spadesuit\}$ satisfying

$$m_\square + m_\clubsuit + m_\diamondsuit + m_\heartsuit + m_\spadesuit = M.$$

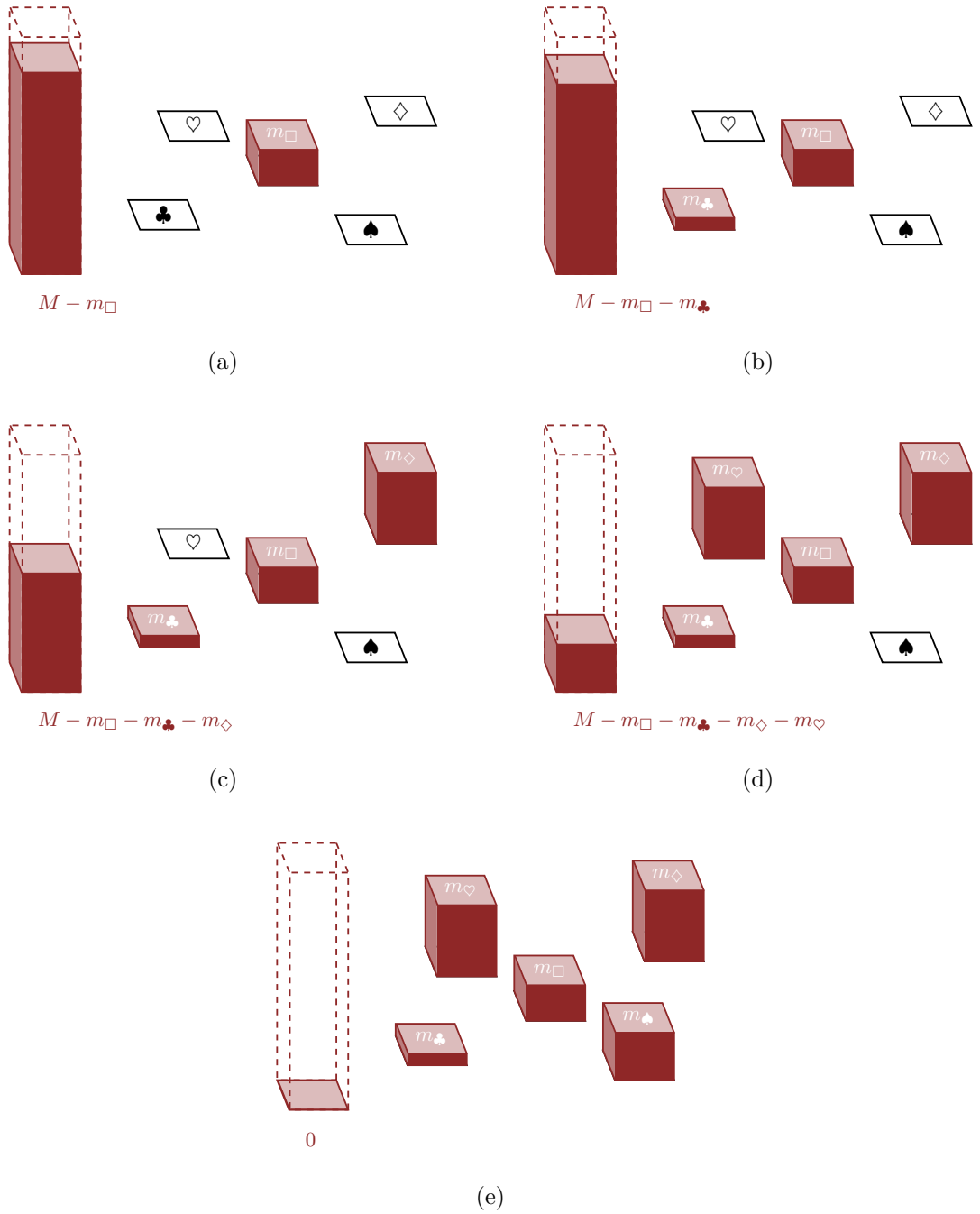


Figure 7: Because the total quantity M is conserved, every allocation m_n to an element $x_n \in X$ depletes the amount available for the allocation to the remaining elements. An exhaustive allocation leaves nothing left in the reservoir after each element has received its allocation.

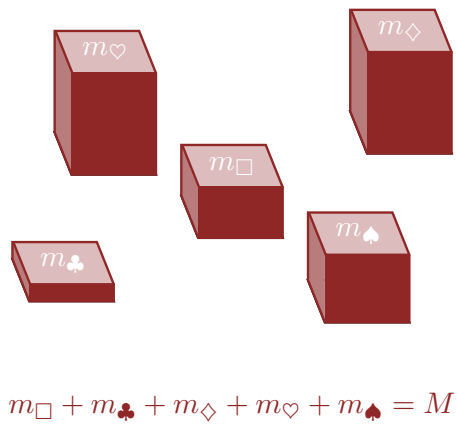


Figure 8: A measure μ over the finite set X is any consistent allocation of M to the elements $x_n \in X$. Every measure can be characterized by N positive real numbers m_n that sum to M, or equivalently a function that maps each element x_n to its allocated measure m_n.

The larger m_n the more of M is allocated to the element x_n. Following this terminology, we will also refer to M as the **total measure** and m_n as the **measure allocated to** x_n.

If the total measure is infinite, $M = \infty$, then *at least one* of the elements in X has to receive an infinite allocation in order to empty the reservoir. That said, we can consistently allocate infinite measure to multiple elements, or even all of the elements, at the same time. Infinite measures can be very generous in their allocations!

The allocation $\{x_n, m_n\}$ defined by a measure μ can also be organized into a function that maps each element to its associated allocation,

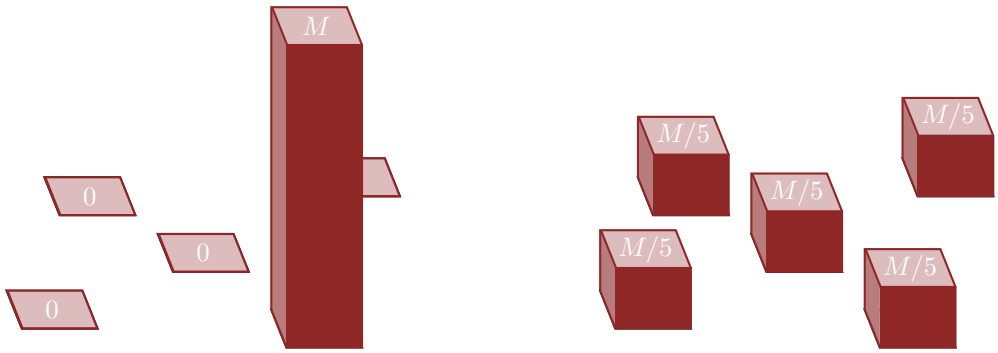
$$\begin{aligned} \mu : X &\to [0, \infty] \\ x_n &\mapsto m_n = \mu(x_n). \end{aligned}$$

A nice conceptual benefit of this functional perspective is that instead of thinking about the global allocation $m_1, \ldots, m_N$ all at once we can isolate local allocations by evaluating μ at only a single input $m_n = \mu(x_n)$.

In general, there are an infinite number of ways to allocate a total measure to the elements of a finite set, and hence an infinite number of measures that we can define over that set. I will denote the collection of all possible measures over X as $\mathcal{M}(X)$.

There are a few exceptional measures within this collection worth mentioning. A **singular measure** allocates the total measure M to a single element, leaving the rest with nothing

(Figure 9a). On the other hand, a **uniform measure** allocates the same measure M/N to each element (Figure 9b). On finite sets, there are N distinct singular measures, one for each distinct element, and a single unique uniform measure.



(a) (b)

Figure 9: A singular measure (a) allocates the total measure to a single element while the uniform measure (b) spreads the total measure to each element evenly.

Perhaps the most important types of measures are those whose total measure is finite, $0 < M < \infty$. Appropriately enough, we refer to these as **finite measures**.

What makes finite measures so special is that we can always reframe the allocation they define into a *relative* one. Instead of considering the absolute measure allocated to each element m_n, we can consider the *proportion* of the total measure allocated to each element, (Figure 10)

$$p_n = m_n / M.$$

By construction, proportions are confined to the unit interval $[0, 1]$. As with any quantity taking values in $[0, 1]$, we can represent proportions equally well with decimals, for example $p_n = 0.2$, and percentages, $p_n = 20\%$.

In other words, a proportional measure defines the function (Figure 11)

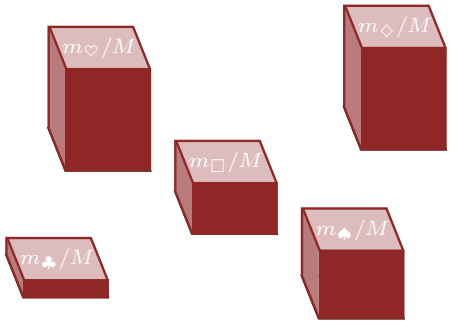
$$\begin{aligned} \pi : X &\to [0, 1] \\ x_n &\mapsto p_n = \pi(x_n) \end{aligned}$$

with

$$0 \leq p_n \leq 1$$

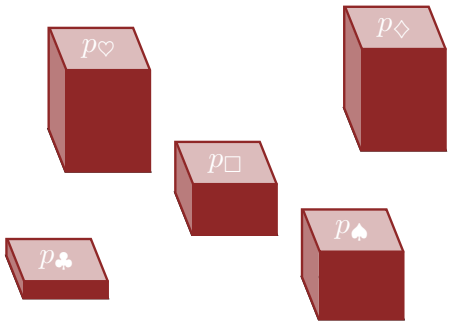
and

$$\sum_{n=1}^{N} p_n = 1.$$



$$\frac{m_\square}{M} + \frac{m_\clubsuit}{M} + \frac{m_\diamondsuit}{M} + \frac{m_\heartsuit}{M} + \frac{m_\spadesuit}{M} = \frac{M}{M} = 1$$

Figure 10: Every finite measure can be characterized by a proportional allocation.



$$p_\square + p_\clubsuit + p_\diamondsuit + p_\heartsuit + p_\spadesuit = 1$$

Figure 11: A proportional allocation is also known as a probability distribution.

A collection of variables $\{p_1, \ldots, p_N\}$ satisfying these properties is referred to as a **simplex**.

A proportional measure π is also known as a **probability distribution**, with the proportional allocations p_n denoted **probabilities**. While the term "probability" is often encumbered with all kinds of interpretational and philosophical baggage, its mathematical structure is really quite straightforward. On a finite set, a probability is just the proportion of some finite quantity that is allocated to an individual element!

Philosophical tension arises only when we try to assign some meaning to these proportional allocations. This isn't a question of mathematics but rather *applying* mathematics to model a system of interest. We'll come back to the many different systems that can be consistently modeled with probability theory, and hence the many different interpretations of probability itself, in Chapter 4.

While we're on the topic of terminological issues, the word "distribution" is not without its own problems. In mathematics, the term "distribution" is heavily overloaded and can be used to refer to a variety of related but formally distinct concepts. One way to avoid confusion is to always refer to proportional measures as "probability distributions" and never just "distributions" alone.

3 Measure and Probability Over Subsets

Regardless of whether is is absolute or proportional, any allocation over the individual elements $x \in X$ of a finite set also defines an allocation over entire subsets $\mathsf{x} \in 2^X$. The measure allocated to a subset is just the sum of the measures allocated to the elements in that subset. For example, the measure allocated to $\mathsf{x} = \{\square, \clubsuit, \heartsuit\}$ is $m_\square + m_\clubsuit + m_\heartsuit$ (Figure 12).

In other words, a measure over individual elements $\mu : X \to [0, \infty]$ implicitly defines a measure over subsets $\mu : 2^X \to [0, \infty]$. Similarly, a probability distribution over individual elements $\pi : X \to [0, 1]$ implicitly defines a probability distribution over subsets $\pi : 2^X \to [0, 1]$.

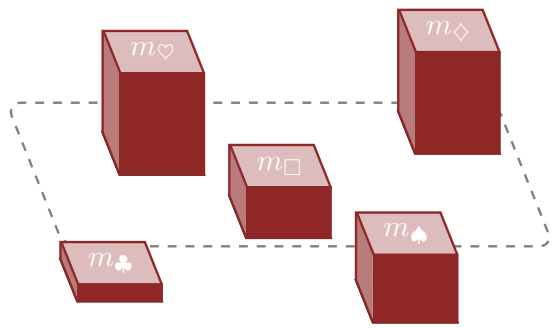
In this last step we have made the cardinal sin of *overloading* our notation so that μ refers to both types of measures. The only way to differentiate them is through their different inputs; $\mu(x)$ denotes the element-wise measure allocated to $x \in X$ and $\mu(\mathsf{x})$ denotes the subset-wise measure allocated to $\mathsf{x} \in 2^X$. Maintaining different typographical conventions for elements and subsets is critical to avoid any confusion when we have overloading notation like this.

By construction, every subset measure and probability distribution satisfies a wealth of useful properties. For instance, for any measure

$$\mu(\emptyset) = 0$$

and

$$\mu(X) = \sum_{n=1}^{N} \mu(x_n) = M.$$



$$\mathsf{x} = \{\square, \diamondsuit, \heartsuit\}$$
$$\mu(\mathsf{x}) = m_\square + m_\diamondsuit + m_\heartsuit$$

Figure 12: On a finite set, an allocation over individual elements also defines an allocation over any subset.

Similarly, for any probability distribution we have $\pi(\emptyset) = 0$ and $\pi(X) = 1$.

Even better, subset allocations play well with the subset operations. Consider, for example, the two disjoint subsets $\mathsf{x}_1 = \{\square, \diamondsuit\}$ and $\mathsf{x}_2 = \{\clubsuit, \spadesuit\}$. Because the two subsets are disjoint, their union simply combines all of their elements,

$$\mathsf{x}_1 \cup \mathsf{x}_2 = \{\square, \diamondsuit\} \cup \{\clubsuit, \spadesuit\} = \{\square, \clubsuit, \diamondsuit, \spadesuit\},$$

and the measure of that union is just the sum of the measures of the input subsets,

$$\begin{aligned}
\mu(\mathsf{x}_1 \cup \mathsf{x}_2) &= \mu(\{\square, \clubsuit, \diamondsuit, \spadesuit\}) \\
&= m_\square + m_\clubsuit + m_\diamondsuit + m_\spadesuit \\
&= (m_\square + m_\diamondsuit) + (m_\clubsuit + m_\spadesuit) \\
&= \mu(\mathsf{x}_1) + \mu(\mathsf{x}_2).
\end{aligned}$$

More generally, for any collection of subsets

$$\mathsf{x}_1, \ldots, \mathsf{x}_K$$

that are mutually disjoint,

$$\mathsf{x}_k \cap \mathsf{x}_{k'} = \emptyset$$

for $k \neq k'$, we have

$$\mu(\cup_{k=1}^K \mathsf{x}_k) = \sum_{k=1}^K \mu(\mathsf{x}_k).$$

In words, if we can decompose a subset into a disjoint collection of smaller subsets then we can also decompose the measure allocated to that initial subset into measures allocated to the component subsets. This consistency property is known as **additivity**.

A subset x and its complement x^c are always disjoint, $\mathsf{x} \cap \mathsf{x}^c = \emptyset$. At the same time, their union covers the entire set, $\mathsf{x} \cup \mathsf{x}^c = X$. Consequently, additivity implies that

$$\begin{aligned} M &= \mu(X) \\ &= \mu(\mathsf{x} \cup \mathsf{x}^c) \\ &= \mu(\mathsf{x}) + \mu(\mathsf{x}^c), \end{aligned}$$

or

$$\mu(\mathsf{x}^c) = M - \mu(\mathsf{x}).$$

In words, the measure allocated to the complement of a subset is the total measure less the measure allocated to that subset. The corresponding result for probability distributions is even cleaner,

$$\pi(\mathsf{x}^c) = 1 - \pi(\mathsf{x}).$$

When two subsets overlap, we have to account for the fact that the sum of their measures $\mu(\mathsf{x}_1) + \mu(\mathsf{x}_2)$ double counts the allocations to common elements. For instance, if $\mathsf{x}_1 = \{\square, \heartsuit\}$ and $\mathsf{x}_2 = \{\square, \spadesuit\}$ then the union includes the overlapping element $\square$ once,

$$\mathsf{x}_1 \cup \mathsf{x}_2 = \{\square, \heartsuit\} \cup \{\square, \spadesuit\} = \{\square, \heartsuit, \spadesuit\}.$$

Consequently, the measure allocated to the union is given by

$$\begin{aligned} \mu(\mathsf{x}_1 \cup \mathsf{x}_2) &= \mu(\{\square, \heartsuit, \spadesuit\}) \\ &= m_\square + m_\heartsuit + m_\spadesuit. \end{aligned}$$

Adding the measures allocated to the two subsets individually, however, includes $m_\square$ twice,

$$\begin{aligned} \mu(\mathsf{x}_1) + \mu(\mathsf{x}_2) &= (m_\square + m_\heartsuit) + (m_\square + m_\spadesuit) \\ &= m_\square + m_\square + m_\heartsuit + m_\spadesuit \\ &= m_\square + \mu(\mathsf{x}_1 \cup \mathsf{x}_2). \end{aligned}$$

Conveniently, the element that is double counted here is exactly the lone element in the intersection of the two initial subsets (Figure 13),

$$m_\square = \mu(\{\square\}) = \mu(\mathsf{x}_1 \cap \mathsf{x}_2).$$

This allows us to write

$$\mu(\mathsf{x}_1) + \mu(\mathsf{x}_2) = \mu(\mathsf{x}_1 \cap \mathsf{x}_2) + \mu(\mathsf{x}_1 \cup \mathsf{x}_2).$$

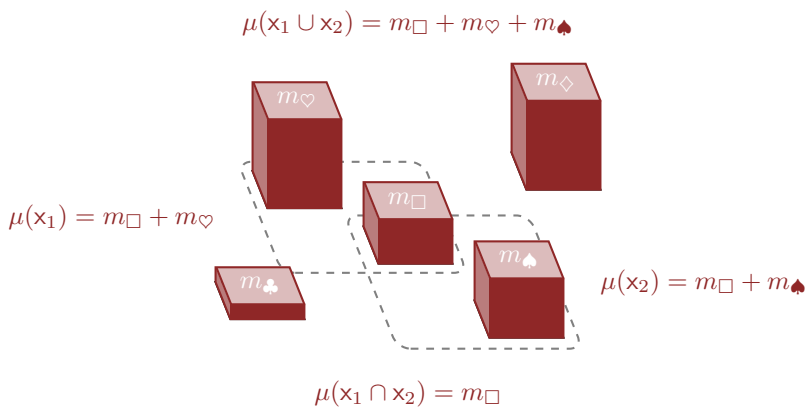


Figure 13: When two subsets overlap, the measure allocated to each double counts the measure allocated to any overlapping elements, here $\square$. The measure allocated to their union, however, does not. This results in an important relationship between the measures allocated to the two subsets, the measure allocated to their union, and the measure allocated to their intersection.

This relationship turns out to be general, holding for *any* two subsets regardless of their overlap.

These subset properties allow us to build up a measure in many different ways, each of which can useful in different circumstances. This provides convenient flexibility when trying to apply measure theory and probability theory in practice.

For example, we can always specify a measure *globally* by specifying the individual allocations at the same time (Figure 14). Alternatively, we can specify the allocation *locally* by allocating measure to each element one at a time (Figure 15).

Critically, we do not always need to start with individual allocations. We can always start by allocating the total measure to disjoint subsets and then iteratively *refining* that allocation to smaller and smaller subsets until we reach the individual elements (Figure 16).

In addition to providing flexible constructions, the subset definition of a measure $\mu : 2^X \to [0, \infty]$ is also important for generalizing measure theory beyond finite sets. Specifically, it becomes *necessary* when trying to consistently define measures on more mathematically complicated sets like the real line. This will be discussed in detail in Chapter 4.

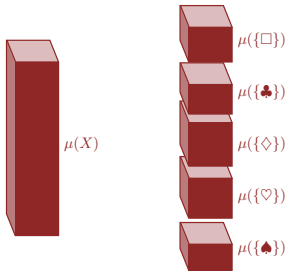


Figure 14: Measures can be constructed by specifying the individual element allocations all at once.

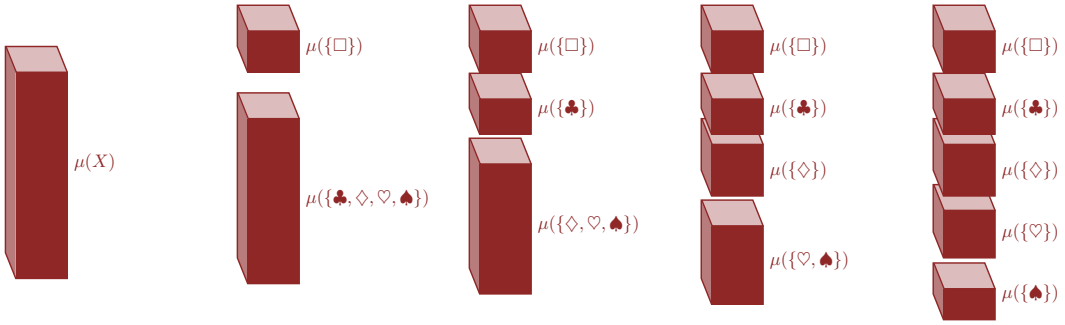


Figure 15: Measures can also be constructed by specifying the individual element allocations one by one.

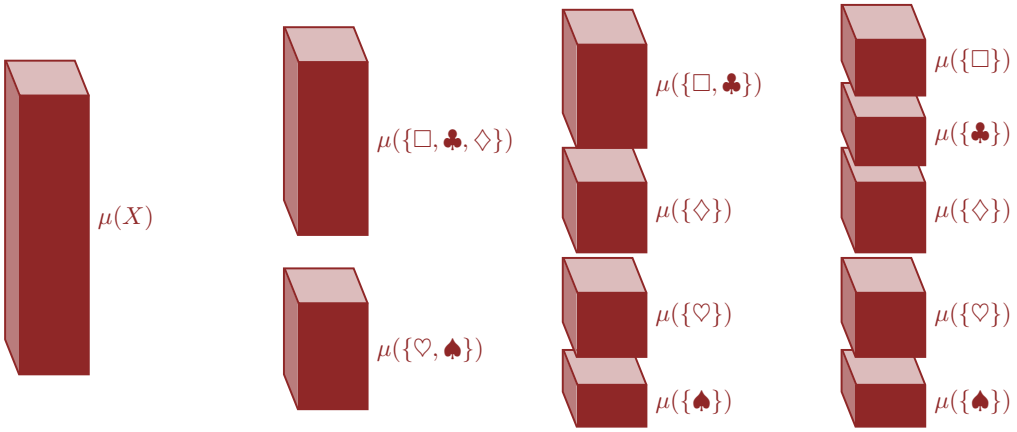


Figure 16: Measures can also be constructed by allocating the total measure to disjoint subsets and then iteratively refining that allocation to smaller and smaller subsets.

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